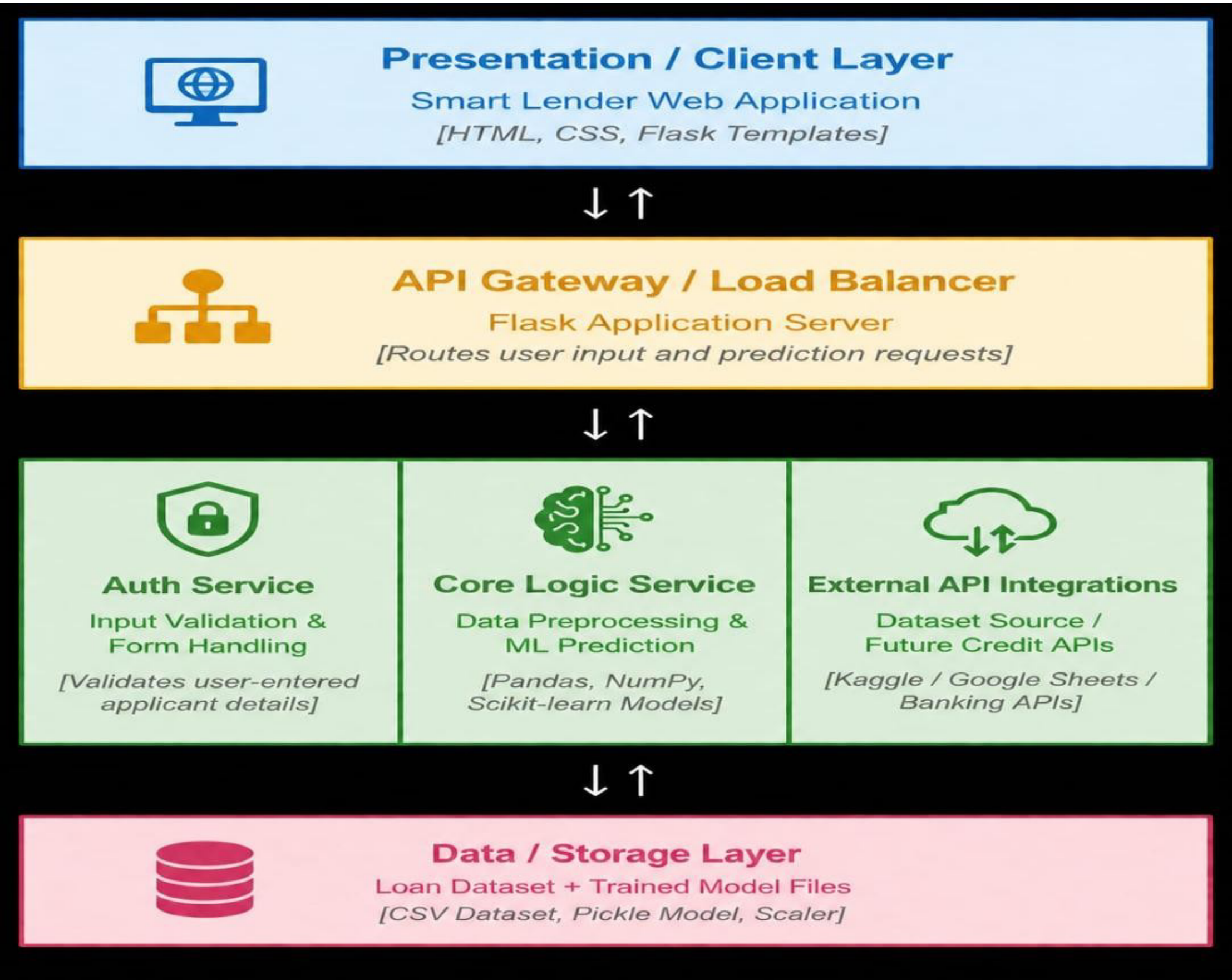


Solution Architecture

Date	27 June 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation layer	Provides a web-based interface where users can enter applicant details such as income, education, loan amount, and credit history to request loan approval prediction.	HTML, CSS, Flask Templates
API Gateway	Acts as the communication bridge between the user interface and the machine learning backend by receiving form data and routing it to the prediction engine.	Flask, Python
Microservices	Handles preprocessing of input data, feature encoding, scaling, model loading, and prediction generation using the trained	Python, Pandas, NumPy, Scikit-learn

	machine learning model.	
Database	Stores the loan prediction dataset, trained machine learning model, scaler, and any additional resources required for deployment and prediction.	CSV Dataset, Pickle Files